

# Proofs of Selected Examples from *The Theory of Partitions*

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# 1 Chapter 1

## 1.1 Example 1

**Proposition.** The number of partitions of  $n$  in which each part appears two, three, or five times equals the number of partitions of  $n$  into parts congruent to 2, 3, 6, 9, or 10 modulo 12.

**Lemma 1.** A positive integer is congruent to 2, 3, 6, 9, or 10 modulo 12 iff it is an odd multiple of 2 or an odd multiple of 3.

*Proof of Lemma 1.* The odd residue classes modulo 12 are  $[1]$ ,  $[3]$ ,  $[5]$ ,  $[7]$ ,  $[9]$ , and  $[11]$ . An odd multiple of 2 has the form  $2m$ , where  $m$  is odd. Thus, the possible residue classes of  $2m$  are

$$\begin{aligned}2[1] &= 2[7] = [2], \\2[3] &= 2[9] = [6], \\2[5] &= 2[11] = [10].\end{aligned}$$

An odd multiple of 3 has the form  $3m$ , where  $m$  is odd. Thus, the possible residue classes of  $3m$  are

$$\begin{aligned}3[1] &= 3[5] = 3[9] = [3], \\3[3] &= 3[7] = 3[11] = [9].\end{aligned}$$

Therefore, the odd multiples of two or three are exactly the integers congruent to 2, 3, 6, 9, or 10 modulo 12.  $\blacksquare$

*Proof of the Proposition.* Let  $p_\alpha(n)$  denote the number of partitions of  $n$  in which each part appears two, three, or five times. Let  $p_\beta(n)$  denote the number of partitions of  $n$  into parts congruent to 2, 3, 6, 9, or 10 modulo 12. Thus

$$\sum_{n \geq 0} p_\alpha(n) q^n = \prod_{n \geq 1} (1 + q^{2n} + q^{3n} + q^{5n}),$$

and by Lemma 1

$$\sum_{n \geq 0} p_\beta(n) q^n = \prod_{n \geq 1} \frac{1}{(1 - q^{2(2n-1)})(1 - q^{3(2n-1)})}.$$

Finally

$$\begin{aligned}
\sum_{n \geq 0} p_\beta(n) q^n &= \prod_{n \geq 1} \frac{1 - q^{4n}}{1 - q^{2n}} \cdot \frac{1 - q^{6n}}{1 - q^{3n}} \\
&= \prod_{n \geq 1} (1 + q^{2n}) (1 + q^{3n}) \\
&= \prod_{n \geq 1} (1 + q^{2n} + q^{3n} + q^{5n}) \\
&= \sum_{n \geq 0} p_\alpha(n) q^n.
\end{aligned}$$

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*Proof of the Proposition (q-Pochhammer).* Let  $p_\alpha(n)$  denote the number of partitions of  $n$  in which each part appears two, three, or five times. Let  $p_\beta(n)$  denote the number of partitions of  $n$  into parts congruent to 2, 3, 6, 9, or 10 modulo 12. Thus

$$\sum_{n \geq 0} p_\alpha(n) q^n = (-q^2; q^2)_\infty (-q^3; q^3)_\infty,$$

and by Lemma 1

$$\sum_{n \geq 0} p_\beta(n) q^n = \frac{1}{(q^2; q^4)_\infty (q^3; q^6)_\infty}.$$

Finally

$$\begin{aligned}
\sum_{n \geq 0} p_\beta(n) q^n &= \frac{(q^4; q^4)_\infty (q^6; q^6)_\infty}{(q^2; q^2)_\infty (q^3; q^3)_\infty} \\
&= (-q^2; q^2)_\infty (-q^3; q^3)_\infty \\
&= \sum_{n \geq 0} p_\alpha(n) q^n.
\end{aligned}$$

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## 1.2 Example 2

**Proposition.** The number of partitions of  $n$  in which only odd parts may be repeated, here denoted  $p_o(n)$ , equals the number of partitions of  $n$  in which no part is repeated more than three times, here denoted  $p_3(n)$ .

*Proof.*

$$\begin{aligned}
\sum_{n \geq 0} p_o(n)q^n &= \prod_{n \geq 1} (1 + q^{2n}) \left( \frac{1}{1 - q^{2n-1}} \right) \\
&= \prod_{n \geq 1} \left( \frac{1 - q^{4n}}{1 - q^{2n}} \right) \left( \frac{1}{1 - q^{2n-1}} \right) \\
&= \frac{\prod_{n \geq 1} (1 - q^{4n})}{\prod_{n \geq 1} (1 - q^{2n})(1 - q^{2n-1})} \\
&= \prod_{n \geq 1} \frac{1 - q^{4n}}{1 - q^n} \\
&= \sum_{n \geq 0} p_3(n)q^n.
\end{aligned}$$

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*Proof (q-Pochhammer).*

$$\begin{aligned}
\sum_{n \geq 0} p_o(n)q^n &= (-q^2; q^2)_\infty (q; q^2)_\infty^{-1} \\
&= \frac{(q^4; q^4)_\infty}{(q^2; q^2)_\infty (q; q^2)_\infty} \\
&= \frac{(q^4; q^4)_\infty}{(q; q)_\infty} \\
&= \sum_{n \geq 0} p_3(n)q^n.
\end{aligned}$$

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### 1.3 Example 3

**Proposition.** The number of partitions of  $n$  in which only parts  $\not\equiv 0 \pmod{2^m}$  may be repeated, here denoted  $p_\alpha(n)$ , equals the number of partitions of  $n$  in which no part appears more than  $2^{m+1} - 1$  times, here denoted  $p_\beta(n)$ .

*Proof.*

$$\begin{aligned}
\sum_{n \geq 0} p_\alpha(n) q^n &= \prod_{n \geq 1} \left(1 + q^{2^m n}\right) \left(\frac{1 - q^{2^m n}}{1 - q^n}\right) \\
&= \prod_{n \geq 1} \left(1 + q^{2^m n}\right) \left(1 + q^n + \cdots + q^{(2^m - 1)n}\right) \\
&= \prod_{n \geq 1} \left(1 + q^n + \cdots + q^{(2^{m+1} - 1)n}\right) \\
&= \prod_{n \geq 1} \frac{1 - q^{2^{m+1}n}}{1 - q^n} \\
&= \sum_{n \geq 0} p_\beta(n) q^n.
\end{aligned}$$

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*Proof (q-Pochhammer).*

$$\begin{aligned}
\sum_{n \geq 0} p_\alpha(n) q^n &= (-q^{2^m}, q^{2^m}; q^{2^m})_\infty (q; q)_\infty^{-1} \\
&= \frac{(q^{2^{m+1}}; q^{2^{m+1}})_\infty}{(q; q)_\infty} \\
&= \sum_{n \geq 0} p_\beta(n) q^n.
\end{aligned}$$

■

## 1.4 Example 4

**Proposition.** The number of partitions of  $n$  with unique smallest part and largest part at most twice the smallest part, here denoted  $p_\alpha(n)$ , equals the number of partitions of  $n$  in which the largest part is odd and the smallest part is larger than half the largest part, here denoted  $p_\beta(n)$ .

### Scratchwork

Let

$$a_n(q) = \frac{q^n}{(q^{n+1}; q)_n}, \quad b_n(q) = \frac{q^{2n-1}}{(q^n; q)_n}, \quad (1)$$

And let

$$A_m(q) = \sum_{n=1}^{2m-1} a_n(q), \quad B_m(q) = \sum_{n=1}^m b_n(q). \quad (2)$$

Then

$$\sum_{n \geq 1} p_\alpha(n) q^n = \sum_{n \geq 1} a_n(q) = \lim_{m \rightarrow \infty} A_m(q) = A(q), \quad (3)$$

And

$$\sum_{n \geq 1} p_\beta(n) q^n = \sum_{n \geq 1} b_n(q) = \lim_{m \rightarrow \infty} B_m(q) = B(q). \quad (4)$$

**Conjecture 1.** For all  $m \in \mathbb{Z}_+$ ,

$$A_m(q) \equiv B_m(q) \pmod{q^{2m}}. \quad (5)$$

Let

$$\rho_r(n, m) = \frac{1}{m} \sum_{j=0}^{m-1} e^{2\pi i j(n-r)/m} = \begin{cases} 1 & \text{if } n \equiv r \pmod{m} \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

so that, with  $x_i \geq 0$ , the coefficient extraction operations are

$$[q^N]a_v(q) = \sum_{x_1+x_2+\dots+x_v=N-v} \prod_{i=1}^v \rho_0(x_i, v+i), \quad (7)$$

And

$$[q^N]b_v(q) = \sum_{x_1+x_2+\dots+x_v=N-(2v-1)} \prod_{i=1}^v \rho_0(x_i, v+i-1). \quad (8)$$

Then we have

$$[q^N]A_m(q) = \sum_{v=1}^{2m-1} [q^N]a_v(q), \quad (9)$$

And

$$[q^N]B_m(q) = \sum_{v=1}^m [q^N]b_v(q). \quad (10)$$

**Conjecture 2.** For all  $0 \leq N < 2m$

$$[q^N]A_m(q) = [q^N]B_m(q). \quad (11)$$

Conjecture 1 and Conjecture 2 are equivalent.

### Other Observations

The ratios between successive terms are

$$\frac{a_{n+1}}{a_n} = q \frac{(q^{n+1}; q)_n}{(q^{n+2}; q)_{n+1}} = q \frac{1 - q^{n+1}}{(1 - q^{2n+1})(1 - q^{2n+2})}, \quad (12)$$

and

$$\frac{b_{n+1}}{b_n} = q^2 \frac{(q^n; q)_n}{(q^{n+1}; q)_{n+1}} = q^2 \frac{1 - q^n}{(1 - q^{2n})(1 - q^{2n+1})}. \quad (13)$$

Also, for  $m > 2$

$$\frac{A_m(q) - B_m(q)}{q^{2m}} = \frac{1 - q^{2\lceil m/2 \rceil + 2}}{1 - q^2} + F(q) \quad (14)$$

Where  $F(q)$  is some power series in which the lowest-degree term is of a degree greater than  $m$ .